

**HOUSE SUBSTITUTE FOR
SENATE BILL NO. 947**

A bill to amend 1994 PA 451, entitled
"Natural resources and environmental protection act,"
by amending sections 31502, 31503, 31504, 31505, 31507, 31508,
31516, and 31518 (MCL 324.31502, 324.31503, 324.31504, 324.31505,
324.31507, 324.31508, 324.31516, and 324.31518), as added by 1995
PA 59, and by adding sections 31509a and 31518a.

THE PEOPLE OF THE STATE OF MICHIGAN ENACT:

- 1 Sec. 31502. (1) ~~"Abandonment" means an affirmative act on the~~
2 ~~part of an owner to discontinue maintenance or operation of a~~
3 ~~dam.~~ **"Additional spillway capacity" means modification to an**
4 **existing spillway or construction of an auxiliary spillway to**
5 **increase flow conveyance capabilities of the dam.**
6 (2) "Administrative procedures act of 1969" means ~~Act No. 306~~



~~of the Public Acts of 1969, being sections 24.201 to 24.328 of the~~
~~Michigan Compiled Laws.~~ **the administrative procedures act of 1969,**
1969 PA 306, MCL 24.201 to 24.328.

(3) "Alteration" means a change in the design of an existing dam that directly affects or may directly affect the structural integrity of a dam.

(4) "Appurtenant works" means the structure or machinery incident to or annexed to a dam that is built to operate and maintain a dam, including ~~spillways,~~ **any of the following:**

(a) **Spillways**, either in a dam or separate from the dam. ~~; low~~

(b) **Low** level outlet works. ~~; and water~~

(c) **Water** conduits such as tunnels, pipelines, or penstocks, located either through the dam or through the abutments of the dam.

(5) "Auxiliary spillway" means a secondary spillway ~~which that~~
~~is operational at all times and does not require stoplog removal or~~
~~gate manipulation.~~ **designed to activate or be operated only during**
large flood flow events.

(6) "Dam" means an artificial barrier, including dikes, embankments, and appurtenant works ~~that impounds,~~ **meets all of**
the following requirements:

(a) **Impounds**, diverts, or is designed to impound or divert water or a combination of water and any other liquid or material in the water. ~~that is~~

(b) **Is**, or will be when complete, 6 feet or more in height. ~~and that has~~

(c) **Has** or will have ~~an~~ **1 of the following:**

(i) **An** impounding capacity at design flood elevation of 5 surface acres or more. ~~Dam~~

(ii) **For a high hazard potential or significant hazard**



1 **potential dam, an impounding capacity at design flood elevation of**
 2 **15 acre-feet or more.**

3 (7) **Dam** does not include a storage or processing tank or
 4 standpipe constructed of steel or concrete, a roadway embankment
 5 not designed to impound water, or a dug pond where there is no
 6 impoundment of water or waste materials containing water at levels
 7 above adjacent natural grade levels.

8 (8) ~~(7)~~—"Days" means calendar days, including Sundays and
 9 holidays.

10 (9) ~~(8)~~—"Design flood" means the design flow rate for spillway
 11 capacity and dam height design.

12 (10) ~~(9)~~—"Design flood elevation" means the maximum flood
 13 elevation that is considered in the design of the spillway capacity
 14 and freeboard for a dam.

15 (11) ~~(10)~~—"Downstream toe elevation" means the elevation of
 16 the lowest point of intersection between the downstream slope of an
 17 earthen embankment and the natural ground.

18 Sec. 31503. (1) "Emergency action plan" means a plan developed
 19 by the owner that establishes procedures for notification of the
 20 department, public off-site authorities, and other agencies of the
 21 emergency actions to be taken ~~prior to~~ **before, during,** and
 22 ~~following~~ **after** an impending or actual failure of a dam.

23 (2) "Enlargement" means any change in or addition to an
 24 existing dam ~~which~~ **that** raises or may raise the design flood
 25 elevation of the water impounded by the dam.

26 (3) "Failed dam" means a dam not capable of impounding water
 27 at its intended level due to a structural deficiency.

28 (4) "Failure" means an incident resulting in an unplanned or
 29 uncontrolled release of water from a dam.



(5) "Federal Energy Regulatory Commission" means the federal agency or a successor federal agency charged with implementing dam safety and licensing requirements for non-federally owned hydropower projects under the federal power act, 16 USC 791a to 825r.

(6) ~~(5)~~—"Flood of record" means the greatest flow rate determined by the department to have occurred at a particular location.

(7) ~~(6)~~—"Freeboard" means the vertical distance between the design flood elevation and the lowest point of the top of the dam.

(8) ~~(7)~~—"Half probable maximum flood" means the largest flood that may reasonably occur over a watershed ~~—~~and is derived from the combination of hydrologic runoff parameters and the half probable maximum ~~storm~~**precipitation** that produces the maximum runoff.

(9) ~~(8)~~—"Half probable maximum ~~storm~~**precipitation**" means the spatial and temporal distribution of the probable maximum precipitation, divided by 2, that produces the maximum volume of precipitation over a watershed.

(10) ~~(9)~~—"Hazard potential classification" means a reference to the potential for loss of life, property damage, and environmental damage in the area downstream of a dam in the event of failure of the dam or appurtenant works.

(11) ~~(10)~~—"Height" means the difference in elevation measured vertically between the natural bed of a stream or watercourse at the downstream toe of the dam, or, if it is not across a stream channel or watercourse, from the lowest elevation of the downstream toe of the dam, to the design flood elevation or to the lowest point of the top of the dam, whichever is less.



(12) ~~(11)~~—"High hazard potential dam" means a dam located in an area where a failure may cause serious damage to inhabited homes, agricultural buildings, campgrounds, recreational facilities, industrial or commercial buildings, public utilities, ~~main highways, or in areas with a population density greater than~~ or equal to 5,000 people per square mile, interstate roads, other freeway roads, other principal arterial roads as defined in the National Functional Classification (NFC) system, class I ~~carrier~~ railroads, ~~or~~ where environmental degradation or natural resource damage would be significant, or where danger to individuals exists with the potential for loss of life.

Sec. 31504. (1) "Impoundment" means the water held back by a dam.

(2) "Inflow design flood" means the flood hydrograph entering the reservoir that is used to design or modify a specific dam and its appurtenant works, including, but not limited to, sizing the spillway and outlet works and evaluating the maximum storage, height of the dam, and freeboard requirements.

(3) ~~(2)~~—"Low hazard potential dam" means a dam located in an area where failure may cause damage limited to agriculture, uninhabited buildings, structures, or ~~township or county~~ minor collector roads, local roads, or nonrated roadways as defined in the National Functional Classification (NFC) system, where environmental degradation would be minimal, and where danger to individuals is slight or nonexistent.

(4) ~~(3)~~—"Maintenance" means the upkeep of a dam and its impoundment and appurtenant works but does not include alterations or repairs.

(5) ~~(4)~~—"One-hundred year flood" means a flood that has a 1%



1 chance of being equaled or exceeded in any given year.

2 (6) ~~(5)~~ "Owner" means a person ~~who~~**that** owns, leases,
3 controls, operates, maintains, manages, or proposes to construct a
4 dam.

5 (7) "Probable maximum flood" means the largest flood that may
6 reasonably occur over a watershed and is derived from the
7 combination of hydrologic runoff parameters and the probable
8 maximum precipitation that produces the maximum runoff.

9 (8) ~~(6)~~ "Probable maximum precipitation" means the
10 theoretically greatest depth of precipitation for a given duration
11 that is physically possible over a given size storm area at a
12 particular geographic location at a certain time of year.

13 Sec. 31505. (1) "Removal" means the physical elimination of a
14 dam or impoundment.

15 (2) "Repair" means to substantially restore a dam to its
16 original condition and includes only ~~such~~ restoration ~~as~~**that** may
17 directly affect the structural integrity of the dam.

18 (3) "Riparian owner" means a person ~~who~~**that** has riparian
19 rights.

20 (4) "Riparian rights" means rights ~~which~~**that** accrue by
21 operation of law to a landowner on the banks of an inland lake or
22 stream.

23 (5) "Significant hazard potential dam" means a dam located in
24 an area where its failure may cause damage limited to isolated
25 inhabited homes, agricultural buildings, structures, ~~secondary~~
26 ~~highways, short line or minor arterial roads or major collector~~
27 **roads as defined in the National Functional Classification (NFC)**
28 **system, railroads not classified as class I, or public utilities in**
29 **areas with a population density less than 5,000 people per square**



1 **mile**, where environmental degradation may be significant, or where
2 danger to individuals exists.

3 (6) "Spillway" means a waterway in or about a dam designed for
4 the discharge of water.

5 (7) "Spillway capacity" means the maximum rate of discharge
6 that will pass through a spillway at design flood elevation.

7 ~~(8) "Two-hundred year flood" means a flood that has a 0.5%~~
8 ~~chance of being equaled or exceeded in any given year.~~

9 Sec. 31507. (1) A person shall not construct, enlarge, repair,
10 reconstruct, alter, **or** remove ~~, or abandon any~~ a dam except in a
11 manner provided for in this part.

12 (2) This section does not apply to maintenance performed on a
13 dam that does not affect the structural integrity of the dam.

14 Sec. 31508. (1) ~~Except as otherwise provided in subsection~~
15 ~~(2), a~~ **A** licensed professional engineer shall prepare all plans and
16 specifications **and the comprehensive evaluation under section**
17 **31518a**, except for minor projects undertaken ~~pursuant to~~ **under**
18 section 31513.

19 ~~(2) A person who is not a licensed professional engineer may~~
20 ~~prepare plans and specifications only for repairs or alterations to~~
21 ~~a dam where the application is made by a nonprofit organization~~
22 ~~under the following circumstances:~~

23 ~~(a) The nonprofit organization has assets of less than~~
24 ~~\$30,000.00, is exempt from taxation under section 501(c)(3) of the~~
25 ~~internal revenue code of 1986, 26 U.S.C. 501, and is not composed~~
26 ~~primarily of the owners of property adjacent to or contiguous to an~~
27 ~~impoundment.~~

28 ~~(b) The proposed repairs or alterations have a projected total~~
29 ~~cost of less than \$25,000.00.~~



1 ~~(c) The impoundment is open to the public and a notice of~~
2 ~~public access is posted.~~

3 (2) Plans and specifications prepared under subsection (1) and
4 a comprehensive evaluation under section 31518a must include
5 documentation that demonstrates that the licensed professional
6 engineer or design team preparing the plans and specifications and
7 comprehensive evaluation has successfully completed or been in
8 consultation with licensed professional engineers who have, at a
9 minimum, completed 5 similar projects, including any of the
10 following project types:

11 (a) Construction of a new dam.

12 (b) Reconstruction of a failed dam.

13 (c) Enlargement of a dam.

14 (d) Complex alteration of a dam, such as significant
15 rehabilitation or reconstruction of an existing dam.

16 (e) Removal of an existing dam.

17 Sec. 31509a. (1) The department shall establish a staggered
18 registration process that meets all of the following conditions:

19 (a) Is divided into 5 rounds and each round lasts not more
20 than 2 years, with the first round of registrations beginning 6
21 months after the effective date of the 2026 amendatory act that
22 added this section.

23 (b) Registers 1/5 of all dams in this state each round.

24 (c) Identifies the dams that are subject to registration in
25 each round based on hazard potential classification and other
26 relevant considerations.

27 (2) Subject to subsection (1), an owner shall register all of
28 its dams with the department on a form and in a manner prescribed
29 by the department. The application for a registration under this



1 section must include all of the following information:

2 (a) Proof of ownership of the dam.

3 (b) An asset management plan for all dams owned by the owner,
4 that includes all of the following information:

5 (i) An inventory of all dam assets.

6 (ii) An assessment of conditions, remaining life, and critical
7 deficiencies and failure modes, and a life cycle cost analysis.

8 (iii) An assessment of probability and consequences of dam
9 failure.

10 (iv) A list of funding sources adequate to cover estimated life
11 cycle costs.

12 (v) A performance evaluation that includes regulatory
13 compliance, inspection schedule, instrumentation monitoring, and
14 internal operations.

15 (c) A demonstrated compliance with this part that includes all
16 of the following:

17 (i) A current inspection report under section 31518 and a plan
18 and schedule approved by the department to address any deficiencies
19 identified in the inspection report, if applicable.

20 (ii) For high hazard potential and significant hazard potential
21 dams, a current, comprehensive evaluation report under section
22 31518a, and a plan and schedule approved by the department to
23 address any deficiencies identified in the comprehensive evaluation
24 report, if applicable.

25 (iii) A current surveillance and monitoring plan under section
26 31518b.

27 (iv) A current operation and maintenance plan under section
28 31518c.

29 (v) A current safety and security plan under section 31518d.



1 (d) Any additional information that the department determines
2 is necessary for dam safety.

3 (3) A registration issued under this section is valid for 10
4 years. Not later than April 1 of the year the registration expires,
5 to renew a registration, an owner shall submit an application for
6 registration under subsection (2).

7 (4) The department shall approve or deny an application for a
8 registration not later than 120 days after the application is
9 complete. The department shall deny an application for registration
10 that does not sufficiently demonstrate that the public health,
11 safety, welfare, property, or natural resources, or the public
12 trust in those natural resources, is protected.

13 (5) The department may revoke a registration, after notice and
14 opportunity for a hearing in accordance with the administrative
15 procedures act of 1969, for any of the following reasons:

16 (a) A violation of the terms of the registration or a change
17 in the condition of the dam.

18 (b) The owner failed to disclose relevant facts or otherwise
19 obtained a registration by misrepresentation.

20 (c) The dam is subject to any of the following:

21 (i) An order under section 31518(7).

22 (ii) An order to limit dam operations or an order for the
23 removal of the dam under section 31519.

24 (iii) An emergency order under section 31521.

25 (iv) Any other violation under section 31524.

26 (6) If the department revokes a registration under subsection
27 (5), the department shall notify the owner in writing of the
28 revocation not later than 30 days after the hearing and state the
29 deficiency and request additional information.



(7) If the department denies a registration under subsection (4) or revokes a registration under subsection (5), the owner shall submit a revised application for registration not later than 180 days after the notification is received under subsection (4) or (6).

(8) An owner can transfer a registration to a new owner if both of the following conditions are met:

(a) The new owner is willing and financially capable of complying with the terms of the registration.

(b) The owner receives written approval from the department for the transfer.

(9) The owner shall submit a request for transfer of a registration under subsection (8) on a form and in a manner prescribed by the department. The request for a transfer of a registration must be accompanied by a written agreement between the owner and new owner that states a specific date for the transfer of responsibility.

Sec. 31516. (1) ~~Spillway~~ Subject to subsection (4), spillway capacity ~~shall~~ **must** meet the following minimum criteria:

(a) Low hazard potential dams ~~shall~~ **must** be capable of passing the 100-year flood, or the flood of record, whichever is greater.

(b) Significant hazard potential dams ~~shall~~ **must** be capable of passing the ~~200-year~~ **half probable maximum** flood, or the flood of record, whichever is greater.

(c) High hazard potential dams ~~, less than 40 feet in height, as measured from the 200-year design flood elevation to the lowest downstream toe elevation, shall be capable of passing the 200-year flood, or the flood of record, whichever is greater.~~

~~(d) High hazard potential dams, 40 feet or greater in height,~~



1 ~~as measured from the 200-year design flood elevation to the lowest~~
 2 ~~downstream toe elevation, shall~~ **must** be capable of passing the half
 3 probable maximum flood, ~~. The half probable maximum flood criterion~~
 4 ~~may be reduced to not less than the 200-year flood, with proper~~
 5 ~~documentation evidencing a failure of a dam under half probable~~
 6 ~~maximum flood conditions will not cause additional flood damage or~~
 7 ~~loss of life.~~ **or the flood of record, whichever is greater.**

8 (d) ~~(e)~~ Spillway design capacity ~~shall not~~ **must** be **not** less
 9 than the flood of record.

10 (2) Freeboard ~~shall~~ **must** be considered when determining
 11 spillway capacity.

12 (3) If a dam cannot pass the design flood, ~~an auxiliary~~
 13 **additional** spillway **capacity** must be provided. The owner must
 14 document, to the satisfaction of the department, that the dam has
 15 sufficient spillway capacity, and that proper means are available
 16 to operate the spillway or spillways during the design flood.

17 (4) **The criteria described under subsection (1) may be reduced**
 18 **in accordance with a site-specific, consequence-based, or risk-**
 19 **based inflow design flood study completed in a manner approved by**
 20 **the department.**

21 Sec. 31518. (1) An owner shall submit to the department
 22 inspection reports prepared by a licensed professional engineer
 23 that evaluate the condition of the dam. The inspection report ~~shall~~
 24 **must** be submitted as follows:

25 (a) Not less than once every ~~3 years~~ **year** for high hazard
 26 potential dams.

27 (b) Not less than once every ~~4~~ **2** years for significant hazard
 28 potential dams.

29 (c) Not less than once every 5 years for low hazard potential



1 dams.

2 (2) The department shall determine the hazard potential
 3 classification ~~of~~ **for** all dams and ~~shall~~ establish an inspection
 4 schedule. The **department shall notify the owner of each dam's**
 5 **hazard potential classification, the effective date of each hazard**
 6 **potential classification, and each dam's hazard potential history.**
 7 **Not later than 30 days after the owner receives the department's**
 8 **notice of a dam's hazard potential classification, the owner shall**
 9 **provide the department with information regarding how the owner**
 10 **will meet the requirements of the dam's hazard potential**
 11 **classification, including, but not limited to, spillway capacity,**
 12 **emergency action plan, inundation mapping requirements, and an**
 13 **inspection schedule. An** inspection schedule ~~shall~~ **developed under**
 14 **this subsection must** require annual submission of inspection
 15 reports for ~~approximately 1/3 of~~ all high hazard potential dams,
 16 ~~1/4~~ **1/2** of all significant hazard potential dams, and 1/5 of all
 17 low hazard potential dams. The department shall notify **the** owners
 18 in writing when inspection reports are due. The department may
 19 order additional inspection reports following an event or change in
 20 condition that could threaten a dam.

21 (3) An inspection report required by this section ~~shall~~ **must**
 22 include, at a minimum, all of the following:

23 (a) An evaluation of the dam's condition, spillway capacity,
 24 operational adequacy, and structural integrity.

25 (b) A determination of whether deficiencies exist that could
 26 lead to the failure of the dam.

27 (c) Recommendations for maintenance, repair, and alterations
 28 of a dam as are necessary to eliminate any deficiencies.

29 (d) **A schedule to address deficiencies that is approved by the**



1 department. An owner violates this part if the owner fails to
2 address deficiencies in accordance with the schedule approved under
3 this subdivision. The department may modify the schedule under this
4 subdivision to address deficiencies based on the conditions of the
5 dam.

6 (e) An assessment of the appropriateness of the current hazard
7 potential classification established under subsection (1) taking
8 into consideration the current conditions that exist downstream of
9 the dam. An assessment under this subdivision must consider
10 conditions related to damage as identified under a significant
11 hazard potential dam classification and serious damage as
12 identified under a high hazard potential dam classification that
13 might have changed since the effective date of the current hazard
14 potential classification established under subsection (1).

15 (f) An assessment of operations and maintenance based on the
16 requirements of the current operation and maintenance records
17 prepared under section 31517a.

18 (g) An evaluation of the risk or changes in risk of
19 significant hazard potential and low hazard potential dams, if
20 applicable.

21 (h) A review of the original design, construction,
22 maintenance, repair, and probable failure mode records of high
23 hazard potential and significant hazard potential dams, if
24 applicable.

25 (i) Recommendations for more detailed investigations or
26 evaluations as needed to determine the condition of the dam.

27 (4) Instead of engaging a licensed professional engineer to
28 prepare an inspection report, local units of government or an
29 organization of the type described in section ~~31508(2)(a) through~~



1 ~~(e)~~-31509(5) may request the department to conduct a visual
 2 inspection of a dam owned by that local unit of government and
 3 prepare a report on the condition of the dam in accordance with
 4 subsection (3). The department shall notify a requesting local unit
 5 of government as to when the inspection is to occur.

6 (5) If an inspection report discloses the need for a more
 7 detailed investigation or evaluation of certain dam features for
 8 the purpose of determining the condition of the dam, the department
 9 may order the completion and submission of that detailed
 10 investigation or evaluation at the expense of the owner. An
 11 investigation or evaluation required under this subsection ~~shall~~
 12 **must** be conducted under the supervision of a licensed professional
 13 engineer.

14 (6) If an owner does not submit an inspection report as
 15 required by subsection (1) or conduct additional investigations if
 16 required by subsection (5), the department or any person ~~who~~**that**
 17 would have life or property threatened by a breach of the dam may
 18 have a report prepared and recover the costs of preparing the
 19 report in a civil action. ~~commenced in a court of competent~~
 20 ~~jurisdiction.~~ This subsection does not limit the right of any
 21 person to bring a cause of action ~~in a court of proper jurisdiction~~
 22 to compel an owner to comply with the requirements of this part.

23 (7) If, based on the findings and recommendations of the
 24 inspection report, ~~and an inspection by the department, or if~~ the
 25 department finds that a condition exists ~~which~~**that** endangers ~~a~~**the**
 26 dam, ~~it~~**the department** shall order the owner to take actions that
 27 the department considers necessary to alleviate the danger.

28 **Sec. 31518a. (1) For high hazard potential and significant**
 29 **hazard potential dams, an owner shall submit to the department a**



1 comprehensive evaluation report prepared by a licensed professional
2 engineer that evaluates the overall safety of the dam. For low
3 hazard potential dams, the department may require an owner to
4 submit a comprehensive evaluation report as provided under this
5 subsection. The comprehensive evaluation report must be submitted
6 not less than once every 10 years on a schedule determined by the
7 department.

8 (2) The comprehensive evaluation report must include, but is
9 not limited to, a detailed evaluation and recommendation for all of
10 the following:

11 (a) The dam's condition, spillway capacity, operational
12 adequacy, and structural integrity.

13 (b) The adequacy of the dam's original design and
14 construction.

15 (c) The adequacy of the dam's operation and maintenance
16 procedures.

17 (d) Probable failure modes and risk reduction measures.

18 (e) Conformance with current dam safety practices.

19 (f) Any deficiencies that exist that could lead to failure of
20 the dam.

21 (g) Maintenance, repair, and alterations of a dam that are
22 necessary to eliminate the deficiencies described under subdivision
23 (f).

24 Enacting section 1. This amendatory act does not take effect
25 unless House Bill No. 5485 of the 103rd Legislature is enacted into
26 law.